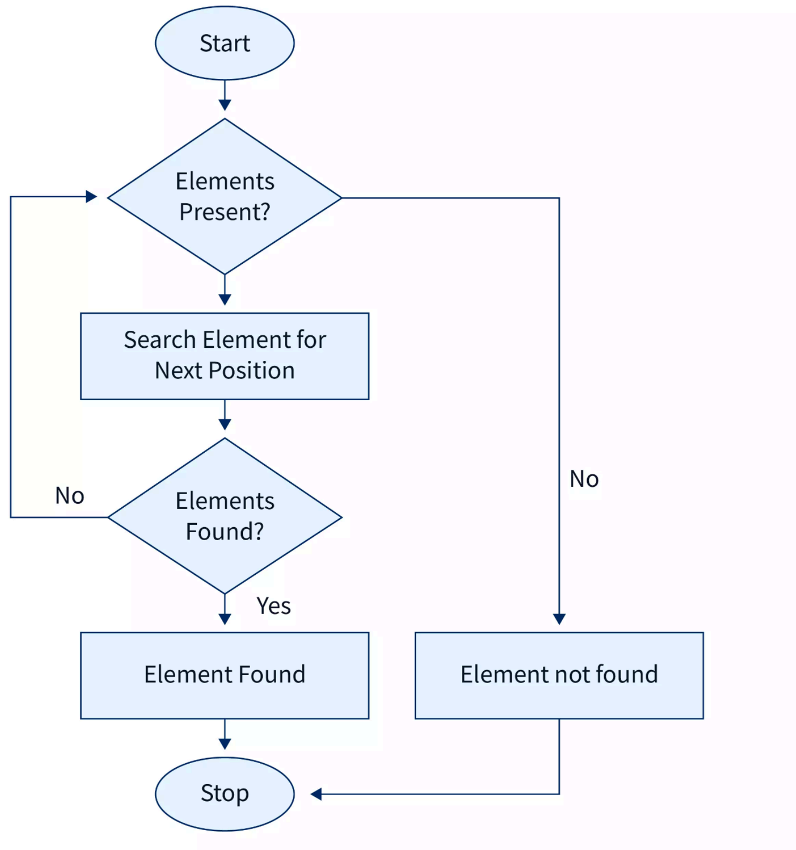
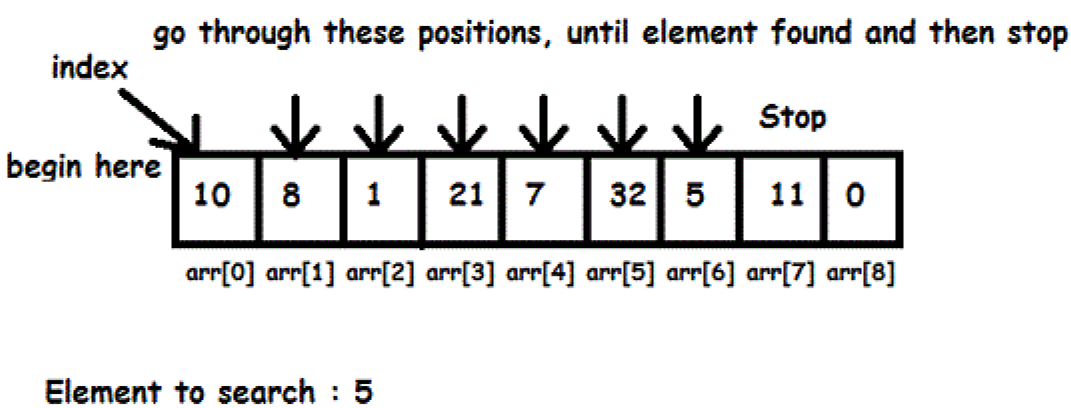
Linear Search Algorithm (C Programming)





1. Introduction to Linear Search

Linear Search is the **simplest searching algorithm** used to find an element in a list or array.

It works by **checking each element one by one from the beginning until the desired element is found or the list ends**.

Example

Array:

[10, 25, 30, 45, 60]

If we search **30**, the algorithm checks:

10 → 25 → 30 ✓

Element found at **position 3**.

2. Linear Search Algorithm Steps

Algorithm to search an element in an array:

1. Start from the **first element** of the array.
2. Compare the **search element** with the current element.
3. If they are **equal**, return the position.
4. If not equal, move to the **next element**.
5. Repeat until the element is found or the array ends.
6. If the element is not found, return **-1** or display "Not Found".

3. Pseudocode of Linear Search

```
LinearSearch(array, n, key)
        for i = 0 to n-1
           if array[i] == key
              return i
        end for

return -1
```

4. C Program for Linear Search

```
#include<stdio.h>

int main()
{
    int arr[100], n, key, i, found = 0;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    printf("Enter array elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");
    scanf("%d", &key);

    for(i = 0; i < n; i++)
    {
        if(arr[i] == key)
        {
            printf("Element found at position %d", i + 1);
            found = 1;
            break;
        }
    }

    if(found == 0)
        printf("Element not found");

    return 0;
}
```

5. Program Explanation

Step 1

Array and variables declared

int arr[100], n, key, i;

Step 2

User inputs array elements.

Step 3

User enters the **search key**.

Step 4

Loop checks each element

for(i = 0; i < n; i++)

Step 5

Comparison

if(arr[i] == key)

If matched → print position.

6. Example Execution

Input

Enter number of elements: 5
Enter elements: 10 20 30 40 50
Enter element to search: 30

Output

Element found at position 3

7. Time Complexity

Case	Complexity
Best Case	O(1)
Average Case	O(n)
Worst Case	O(n)

Explanation:

- **Best case:** element found at first position
- **Worst case:** element found at last position or not found

8. Advantages of Linear Search

✔ Very **simple algorithm**
✔ Works on **unsorted arrays**
✔ Easy to implement

9. Disadvantages of Linear Search

Slow for **large datasets**
Checks elements **one by one**

10. When Linear Search is Used

Linear search is useful when:

- Data size is **small**
- Array is **unsorted**
- Simplicity is require